

Flow Sequence-Based Anonymity Network Traffic Identification with Residual Graph Convolutional Networks

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Abstract—Identifying anonymity services from network traffic is a crucial task for network management and security. Currently, some works based on deep learning have achieved excellent performance for traffic analysis, especially those based on flow sequence (FS), which utilizes information and features of the traffic flow. However, these models still face a serious challenge because of lacking a mechanism to take into account relationships between flows, resulting in mistakenly recognizing irrelevant flows in FS as clues for identifying traffic. In this paper, we propose a novel FS-based anonymity network traffic identification framework to tackle this problem, which leverages Residual Graph Convolutional Network (ResGCN) to exploit relationships between flows for FS feature extraction. Moreover, we design a practical scheme to preprocess the raw data of real-world traffic, which further improves identification performance and efficiency. Experimental results on two real-world traffic datasets demonstrate that our method outperforms state-of-the-art methods by a large margin.

Index Terms—Anonymity network, traffic classification, flow sequence, deep learning, graph convolutional networks.

I. INTRODUCTION

Due to the increasing demand for the protection of personal network meta-data, anonymity networks have grown in popularity. They provide a way for users to achieve anonymity online. In detail, anonymity networks use encryption and obfuscation methods in communication to conceal the transmission content and the true identity of users. However, while making communication harder to trace and identify, anonymity networks also provide hiding space for illegal and criminal activities [1]–[3]. Thus, it is essential to design an effective traffic analysis system to supervise anonymity networks.

In the early stage, the traditional network traffic analysis system, mainly based on rules to realize scene identification, plays a vital role in network monitoring. These methods identify a limited number of services by analyzing the inherent components (i.e., port numbers and characteristic values) in the traffic packet. Unfortunately, the identification method based on port numbers has become invalid due to the mechanism of anonymity networks and many other networks using random ports [4]. At the same time, the growth of network scale and the application of encryption technology have also adversely affected the performance of traffic analysis methods based on characteristic values.

Recently, benefiting from the development of artificial intelligence (AI) technology, many efforts [5]–[12] have been

made to develop various traffic analysis methods based on machine learning (ML) and deep learning (DL) algorithms. The AI-based traffic analysis methods monitor traffic data's basic and statistical characteristics to better deal with complex and changeable network traffic. There is no doubt that AI technology has greatly improved the performance of traffic analysis methods. In addition, we note that many studies have shown that the traffic data has obvious spatio-temporal correlations [13]–[15]. For example, when a user uses an anonymity network to browse the web, many related application requests will appear in a short period. These application requests will generate corresponding multiple flows in the traffic. Based on the spatio-temporal correlation, some researchers proposed methods [16]–[18] to establish flow sequences to achieve better traffic analysis performance. However, they all ignore some critical relationships between flows, resulting in mistakenly taking irrelevant flows in the flow sequence as clues for traffic identification. Therefore, to make the identification more accurate, we should leverage these relationships to extract features from flow sequences. The definitions of these relationships are listed below.

- **Attribute Relationship.** The same application request generated forward direction flow (F-flow) and corresponding reverse direction flow (R-flow) have the attribute relationship.
- **Time Relationship.** The time relationship represents the time interval between the flows. The longer the interval, the lower the correlation between them.

Due to the structural constraints of current methods implemented based on DL algorithms (e.g., CNN and LSTM), they cannot consider the two relationships during the feature extraction process. So how can we leverage these relationships? The graph convolutional network (GCN), which connects neighbor nodes through edges with different weights and updates the feature representation by calculating the feature summation of neighbor nodes, provides a way to solve this problem by connecting flows according to their relationships.

In this paper, we propose a novel flow sequence-based traffic identification framework with residual graph convolutional network (ResGCN). We take continuous multiple flows (i.e., the flow sequence) as the input feature. Each flow is a node, and the weight of the edge is set according to the relationships between flows. To better preserve the spatial structure of the

gradient, we adopt the residual structure to extract features. Besides, as an end-to-end framework for identifying real-world traffic, it extracts rich features (e.g., statistical features, basic features, etc.) from the raw traffic according to the flow segmentation scheme, and uses the Light Gradient Boosting Machine (LightGBM) algorithm to select the optimal combination of features, improving the model performance and efficiency.

In summary, this paper makes the following contributions:

- We propose exploiting the attribute and time relationships between the flows and realizing a more reasonable and effective feature extraction of the flow sequence for traffic identification. We posit that graph convolutional network (GCN) is suitable for our purpose, and propose a novel RESGCN model to identify different network services. To our best knowledge, this is the first investigation in this direction.
- We design a practical scheme to process real-world raw traffic data. It considers flow segmentation, which is used to generate and enrich the flow features of the raw traffic, and LightGBM-based feature combination, which avoids the insignificant features from reducing model performance and efficiency.
- We evaluate the framework on two real-world traffic datasets. Experimental results show that our method has a more excellent classification performance than other methods, and it is suitable for identifying different network services.

II. RELATED WORK

The identification of anonymity network traffic is a typical traffic classification task. Network traffic classification methods are mainly implemented using three methods (i.e., rules, ML, and DL). The rule-based analysis method can only analyze the uncomplicated network traffic, and the port number identification has been proved to be invalid [19] in the complex network environment.

To solve the issues with rule-based analysis methods, researchers applied ML and DL algorithms to analyze the traffic of anonymity networks or other networks. For instance, Cai *et al.* [2] adopted feature selection algorithm to filter out some irrelevant and redundant features, and then used XGBoost-based algorithm to identify Tor, I2P, and JonDonym networks for four scenarios; Wang *et al.* [9] proposed an encrypted traffic classification method based on 2D-CNN, which converted traffic data into image representations; Lin *et al.* [10] proposed an intrusion detection system with stacked sparse auto-encoder (SSAE) and recurrent neural network (RNN), which extracts traffic features with the greedy layer-wise strategy to obtain good detection performance. Due to the structural constraints of aforementioned algorithms, they cannot consider the relationships between flows during the feature extraction process. Recently, we have also noticed some works exploring graph neural networks (GNN) for traffic classification tasks. Shen *et al.* [11] proposed to use different applications' graph structures (e.g., spindle-shaped, fish-shaped) as input features

for classification. Obviously, a large amount of traffic packets are needed to form structures with different characteristics. Sun *et al.* [12] designed a traffic classifier based on GNN and k-nearest neighbors. Their edge weights of the graph are the similarity of 5 nearest neighbors selected by k-nearest neighbor (KNN). However, the graph of this method relies heavily on the KNN, which means that the quality of the graph is not stable.

To conquer the limitations in current studies, we propose a RESGCN classifier for anonymity network traffic identification, which explores the use of attribute relationships and time relationships between flows to build graph structures for feature extraction. Our method uses statistical features of flows to counter obfuscation and encryption techniques (as described in Section III-A), and builds graphs in only eight continuous flows for real-time analysis.

III. PROBLEM FORMULATION AND APPROACH OVERVIEW

In this section, we first provide the background of flows in the traffic. Then, we introduce the preliminaries of ML-based feature selection algorithm and DL-based feature extraction algorithm. Finally, we present an overview of our approach to identifying the various anonymity services in the network traffic.

A. Flow Generation

In packet switching networks, traffic flow is a sequence of packets carrying information between two hosts. The raw traffic data is captured and stored in pcap (abbreviation of packet capture) format. As a kind of encrypted traffic, it is more difficult to identify different services by directly analyzing the pcap file of anonymity network traffic. Thus, it is necessary to calculate and extract the statistical features to enrich the feature information of each flow.

The statistical features of encrypted traffic are calculated through the *Layer 2* to *Layer 4* header features, which are not affected by any encryption on *Layer 7*¹. These statistical features reflect the characteristics of the traffic from many aspects. For instance, the inter-arrival time (IAT) represents the packet interval; the maximum, minimum, and average package size of the *Layer 3* reflects the package size characteristics of the traffic; packet length (PL) represents the length of the packet produced by each flow from the beginning. Note that as above-mentioned are only a tiny part of the statistical features. Since the statistical features are computed using the lightweight traffic feature extraction tool *Tranalyzer*², which is not the contribution of our method, we only briefly introduce them.

To preserve the spatio-temporal correlations of the raw traffic, we need to store the generated flow sequentially according to the timestamp during extraction. As shown in Fig. 1, we take continuous I flows as a flow sequence, where each flow contains a total of J features. The raw traffic is composed of K

¹In the OSI reference model, the communications between a computing system are split into seven different abstraction layers, labelled 1 to 7.

²<https://tranalyzer.com>

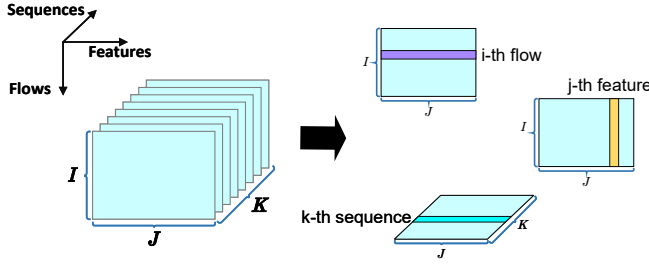


Fig. 1. The traffic flow model.

flow sequences, and each sequence is continuous in time with the previous sequence. As the same type of network behavior often lasts for a certain period, building a flow sequence for analysis has become an effective method to improve classification performance. Obviously, there are some relationships (i.e., attribute and time relationships) between flows in each sequence. If these relationships can be considered in the feature extraction process, the classifier network's design will be more reasonable and effective. Unfortunately, there is currently no related work that combines these relationships for feature extraction.

In addition, many variables will affect the results generated during the flow generation process. First, flows can be segmented according to different durations or packet sizes, directly affecting the calculation results of related statistics. Second, the generated flows contain different statistical features, but some may be meaningless for traffic identification. To achieve optimal flow generation results, we should consider the following two aspects:

- **Effectiveness:** Effectiveness means that the classifier can achieve excellent classification performance by using the generated flows.
- **Timeliness:** Timeliness means that anonymous services can be identified as quickly as possible.

Therefore, based on effectiveness and timeliness, the performance of different segmentation methods and selected features will be comprehensively evaluated in Section V-C.

B. LightGBM for Feature Selection

Feature selection is to remove insignificant features in the generated flows, which can make the generated results more in line with effectiveness and timeliness [20]–[22]. In terms of effectiveness, insignificant features can be characterized as noise under certain conditions, which can adversely affect the identification result. In terms of timeliness, removing these features can reduce the calculation of statistics in the flow generation process and speed up flow generation. In addition, low-dimensional features will also reduce the complexity and classification time of the feature extraction network.

LightGBM is an ML algorithm based on a gradient boosting decision tree (GBDT). As the GBDT algorithm sorts the importance of features during the training process, it is very suitable for feature selection tasks. Traditional GBDT-based algorithms (e.g., PGBRT and XGBoost) have achieved good

performance. However, these methods are time-consuming since they have to scan all sample points for each feature to select the best segmentation point. LightGBM greatly reduces the time complexity of processing samples through the gradient-based one-side sampling (GOSS) algorithm [23]. The main idea of the GOSS algorithm is that samples with large gradients play a major role in the calculation of information gain, which means that these samples with large gradients will contribute more information gain. Therefore, to maintain the accuracy of information gain evaluation, samples with large gradients can be retained when samples are down-sampled, and samples with small gradients are randomly sampled in proportion. Due to the reduction of a large number of data samples with small gradients, the amount of calculation is greatly reduced.

C. GCN for Feature Extraction

Graph convolution network (GCN) is derived from graph spectral theory [24], which extends the convolution operation from grid-based data to graph structure data. GCN also solved natural language processing (NLP) tasks with similar issues (i.e., the relationship between words in a sentence) and achieved great success [25]. In this research, we regard the flow sequence as a graph, and each flow is a node in the graph. A connection relationship is formed according to the relationship between different flows.

Based on the decomposition of graph Laplacian matrix, the signal x on the graph G is filtered by a kernel g_θ :

$$g_\theta *_{\mathcal{G}} x = g_\theta(\mathbf{L})x = g_\theta(\mathbf{U}\mathbf{\Lambda}\mathbf{U}^T)x = \mathbf{U}g_\theta(\mathbf{\Lambda})\mathbf{U}^T x, \quad (1)$$

where $*_{\mathcal{G}}$ denotes a graph convolution operation. $\mathbf{L}$ is the graph Laplacian matrix. The eigenvalue decomposition of the Laplacian matrix is $\mathbf{L} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^T$, where $\mathbf{\Lambda}$ is a diagonal matrix, and $\mathbf{U}$ is Fourier basis.

However, the decomposition becomes very difficult when the scale of the graph is large. For this purpose, Chebyshev polynomials are adopted to solve this computational problem:

$$g_\theta *_{\mathcal{G}} x = g_\theta(\mathbf{L})x = \sum_{m=0}^{M-1} \theta_m C_m(\tilde{\mathbf{L}})x, \quad (2)$$

where θ_m is the learnable parameters. $\tilde{\mathbf{L}} = \frac{2}{\lambda_{max}}\mathbf{L} - \mathbf{I}_N$, in which λ_{max} is the maximum eigenvalue of the Laplacian matrix and $\mathbf{I}_N$ is a unit matrix. The recursive definition of the Chebyshev polynomial is $C_m(\tilde{\mathbf{L}}) = 2\tilde{\mathbf{L}}C_{m-1}(\tilde{\mathbf{L}}) - C_{m-2}(\tilde{\mathbf{L}})$ and $C_1(\tilde{\mathbf{L}}) = \tilde{\mathbf{L}}, C_0(\tilde{\mathbf{L}}) = \mathbf{I}_N$.

D. Our Approach

In this paper, our research focuses on how to identify different anonymity services in real-world anonymity networks more effectively and quickly. Thus, our approach includes a novel traffic classifier and the scheme of traffic acquisition, flow generation, and feature selection.

Our approach includes three steps. As the first step, we configure the switch mirror port, and use the traffic capture tool *tcpdump* to obtain the real-time traffic and save it as a series of pcap files. The frequency of traffic acquisition can

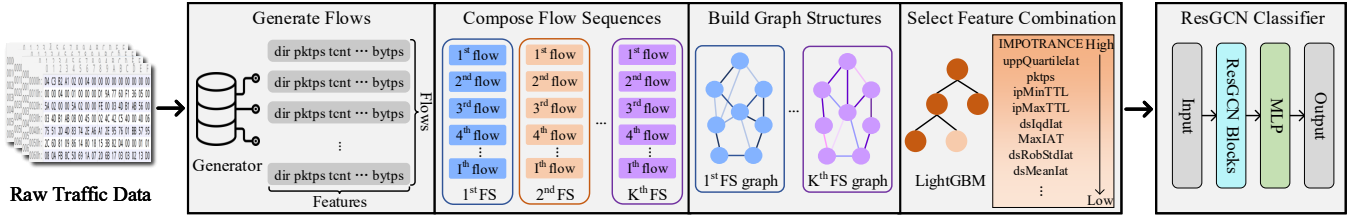


Fig. 2. Overview of the flow sequence-based anonymity network traffic identification framework.

be set differently according to the actual situation. Second, we use a flow generator to realize the fast and real-time analysis of pcap files. According to the preset rules, we extract the flow in the pcap file. The flow sequence is composed of multiple continuous flows, and the relationship graph between different flows will also be generated in this step. Then, the LightGBM-based feature selection method is used to select the optimal feature combination. Finally, the proposed RESGCN classifier leverages generated relationship graphs to realize effective anonymity network traffic identification.

IV. FRAMEWORK OF ANONYMITY NETWORK TRAFFIC IDENTIFICATION

In this section, we introduce our anonymity network traffic identification method. After processing the raw traffic data, RESGCN is adopted to realize the anonymity services identification. The overview of the proposed method is shown in Fig. 2, and the process of our approach is summarized in **Algorithm 1**. The details of each processing stage in the proposed framework are presented as follows.

Algorithm 1 Pseudocode of the Proposed Framework

Input: raw traffic data D_r , total epoch times n
Output: ResGCN model for anonymity network traffic identification

- 1: **procedure** Data_Preprocessing (D_r)
- 2: Generate flows and compute the standardization result
- 3: Take continuous multiple flows as flow sequences
- 4: Build relationship graph between flows
- 5: Use LightGBM-based method to select feature combination
- 6: $D_p \leftarrow$ New training dataset after preprocessing
- 7: $G_p \leftarrow$ Graph structure of the training data
- 8: **return** D_p, G_p
- 9: **procedure** ResGCN_Model (D_p, G_p)
- 10: **while** $i \leq n$ **do**
- 11: Load the proposed network
- 12: Input D_p, G_p into the network for training
- 13: Update the model according to cross entropy loss
- 14: Save $ResGCN_i$ model
- 15: **end while**
- 16: Save $ResGCN_n$ as the final model

A. Raw Traffic Data Processing Scheme

As mentioned in Section III-A, the raw traffic data is obtained as the pcap file, which needs to be processed to be more effectively applied to the RESGCN model. Data processing includes the following four steps.

1) **Generate Flows:** In real-world traffic, e.g., the service provided by *BitTorrent* will cause a very long flow duration. If we do not segment the long-duration flow, it will adversely affect the classification efficiency. Flow segmentation schemes can be divided into two categories: time-based and size-based. The time-based segmentation scheme sets the upper limit of the duration of the flow, and the size-based segmentation scheme sets the upper limit of the maximum packet size. Thus, we first use the generator to generate rich features from the raw traffic through a time-based or size-based flow segmentation scheme. Then, the following standard normalization method is adopted to improve the reliability of the data:

$$z = \frac{x - \mu}{\sigma}, \quad (3)$$

where μ represents the mean of the original data, and σ the standard deviation of the original data.

2) **Compose Flow Sequences:** Some previous studies on traffic classification have shown that the use of flow sequences will significantly improve the classification performance [16]–[18]. However, a flow sequence containing too few flows has insufficient information and cannot achieve the ideal classification performance; too many flows will bring more computational burden and reduce efficiency. Thus, we set each of the flow sequence to contain eight continuous flows.

3) **Build Graph Structures:** Graph generation is the key to the successful application of GCN. We want to achieve a more effective and reasonable analysis of the flow sequences through the graph structure. Obviously, there are many relationships between the different flows in each sequence. We build the graph structure from the following two aspects.

Attribute Relationship Graph (ARG): As mentioned in Section I, the relationship between the F-flow and the R-flow generated by an application request is defined as the attribute relationship. Obviously, these two flows have a strong correlation. We determine the attribute relationship between the two flows through the 3-tuple (i.e., flow index, transmitted bytes, and received bytes). During the flow generation process, a flow index is generated according to the source/destination IP, source/destination port, and protocol. However, the segmentation scheme splits the flow, resulting in a situation where

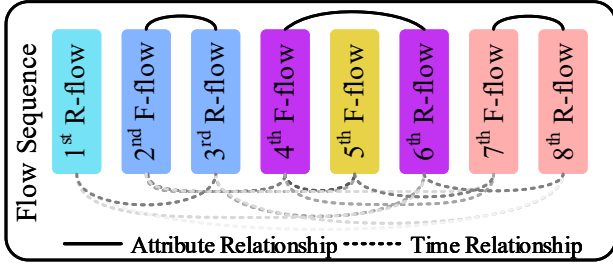


Fig. 3. The time relationship and attribute relationship of different flows in the flow sequence.

there are multiple continuous flows with the same index. if the two flows have the same flow index and swap the number of transmitted bytes and received bytes, they have an attribute relationship. Based on the 3-tuple, we connect the F-flow and the R-flow and set their attribute relationship weight to 1.

$$G_a(V, E) = 3TupleMatching. \quad (4)$$

Time Relationship Graph (TRG): The continuous multiple flows are arranged in order according to the generation time of the flow. We build TRG for different flows based on generation time. If the generation time of the traffic is closer, the higher the weight will be set. Specifically, in each flow sequence, the A-th flow is $flow_a$, and the B-th flow is $flow_b$. The distance between the two flows is $|B - A|$, and the initial weight is $\frac{1}{|B-A|}$. Only flows in the same direction (i.e., forward direction or reverse direction) will be connected.

$$G_t(V, E) = distance^{-1}. \quad (5)$$

To set the time relationship weights of each flow more reasonably, we input the time relationship weights of each flow into the Softmax function, so that the sum of the new weights is 1. Suppose we have n time relationship weights in a flow, we can denote the process as:

$$w'_1, w'_2, \dots, w'_n = Softmax(w_1, w_2, \dots, w_n). \quad (6)$$

As shown in Fig. 3, different flows in the flow sequence are connected through the attribute relationship and time relationship. After establishing the ARG and TRG, we perform normalized on the adjacency matrix of these two graphs to obtain the fusion graph.

4) Select Feature Combination: The LightGBM-based feature selection method is designed to select the optimal feature combination, which can effectively avoid the insignificant features from reducing the performance and efficiency of the classifier.

To evaluate the importance of the j -th feature shown in Fig. 1, we adopt LightGBM to calculate the gradient of each sample. For the gradient calculation of each sample, O is the flows on a fixed node in the decision tree. The variance gain of node split feature j at point d is expressed as:

$$V_{j|O}(d) = \frac{1}{n_O} \left(\frac{(\sum_{x_i \in O: x_i \leq d} g_i)^2}{n_{l|O}^j(d)} + \frac{(\sum_{x_i \in O: x_i > d} g_i)^2}{n_{r|O}^j(d)} \right) \quad (7)$$

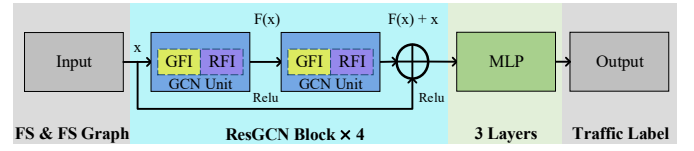


Fig. 4. The structure of ResGCN.

where $n_O = \sum I[x_i \in O]$, $n_{l|O}^j(d) = \sum I[x_i \in O : x_i \leq d]$ and $n_{r|O}^j(d) = \sum I[x_i \in O : x_i > d]$. Traverse the split point of each feature, find the split point $d_j^* = \operatorname{argmax}_d V_j(d)$ and calculate the maximum information gain $V_j(d_j^*)$, and then divide the data into the left and right child nodes according to the split point d_j^* of feature j^* . After sorting and random sampling operations, the information gain is estimated by the following formulas:

$$v_l = \sum_{x_i \in A_l} g_i + \frac{1-a}{b} \sum_{x_i \in B_l} g_i, \quad (8)$$

$$v_r = \sum_{x_i \in A_r} g_i + \frac{1-a}{b} \sum_{x_i \in B_r} g_i, \quad (9)$$

$$\tilde{V}_j(d) = \frac{1}{n} \left(\frac{v_l^2}{n_{l|O}^j(d)} + \frac{v_r^2}{n_{r|O}^j(d)} \right). \quad (10)$$

In subsequent experiments, different comparison algorithms will be used to implement feature selection, and the performance of these algorithms will be compared.

B. Flow Sequence-based RESGCN Classifier

Some previous studies have considered the use of flow sequences to achieve better traffic classification performance [16]–[18], but they cannot carry out directed information exchange between flows. Aiming to address the limitations of these methods, we leverage a GCN over the fused graph of the flows, which allows related flows in the flow sequence to exchange information.

The input of RESGCN is a flow sequence consisting of eight continuous flows after feature selection, which can be constructed as a graph mentioned above. Every flow is seen as a node in a graph. The entire model is composed of four ResGCN blocks and a 3-layer MLP.

The structure of the ResGCN block is shown in Fig. 4. It consists of two GCN units, and the activation function ReLU is used in between. The residual structure is applied in the ResGCN block, which can solve the degradation problem that occurs as the depth of the network increases. The GCN unit includes two key components, i.e., the generated features interaction module (GFI) and related flows interaction module (RFI). The GFI is a fully connected layer without bias. It performs a linear transformation on the features (e.g., packet interval, package size, etc.) of each flow, allowing different features to interact. The RFI enables related flows to exchange information based on the relationship graph. According to the fusion graph mentioned above, these two modules can perform effective information interaction on different features and related flows.

TABLE I
SUMMARY OF THE DATASETS USED FOR EVALUATION.

Dataset	Category	Training dataset	Test dataset
SJTU-AN21 (D ₁)	Eepsites	1,825	1,197
	IRC	3,009	484
	Snark	7,284	1,784
	Video	12,577	581
	JonDonym	1,254	967
	Bittorrent	198	76
	Chat	624	150
	FTP	364	184
	Streaming	949	598
	Browsing	1,130	958
	Total	29,214	6,979
ISCXVPN2016 (D ₂)	Browsing	3,646	912
	FTP	4,446	1,112
	P2P	736	184
	VoIP	8,464	2,117
	Email	689	173
	Streaming	1,523	381
	Chat	3,209	803
	Total	22,713	5,682

After information interaction through 4 ResGCN blocks, a dropout layer is used to improve model generalization ability and reduce overfitting. Finally, the output of the dropout layer is flattened, and a 3-layer MLP with two hidden layers and one output layer is used to perform traffic classification. The first hidden layer consists of a linear layer with output size 220, followed by rectified linear units (ReLU) [26]. The second hidden layer has a similar structure but with an output size of 110. The hidden layer can learn nonlinear functions for feature extraction.

V. PERFORMANCE EVALUATIONS

In this section, we present and discuss our experimental results. Especially, we answer the following three research questions:

- **RQ1:** How effective is the traffic data processing scheme in processing real-world raw traffic? (Section V-C)
- **RQ2:** How well does RESGCN work on real-world network traffic, identifying different network services? (Section V-D)
- **RQ3:** If RESGCN achieves better performance than the state-of-the-art methods? (Section V-E)

A. Datasets for Evaluation

Currently, there are two mainstream datasets (i.e., *ISCX-Tor2016* [27] and *Anon17* [28]) for anonymity network traffic analysis, released in 2016 and 2017, respectively. Due to many changes in protocols and communication mechanisms of anonymity networks over the years, these two datasets are no longer suitable for the current anonymity network traffic analysis. Thus, we use the latest SJTU-AN21 dataset (D₁), which includes the traffic data in the latest version of the three most popular anonymity networks (i.e., Tor, I2P, JonDonym)

generated by ten anonymity services³. There are 29,214 flows and 6,979 flows in the training and test datasets, respectively.

To evaluate the generality of the classifier, we introduce another real-world traffic dataset ISCXVPN2016 (D₂) [29]. This dataset provides the traffic of different application services over virtual private network (VPN). VPN also provide anonymity by creating a private network from a public internet connection. Unlike Tor, I2P, and JonDonym, which are slow and privacy-oriented, VPN is faster and more suitable for daily tasks like casual browsing and streaming. A total of 7 application services (Browsing, FTP, VoIP, etc.) are included in the selected dataset, and 28,395 flows are parsed for classification. The ISCXVPN2016 dataset is divided into an independent training dataset and test dataset according to the proportion of 80% and 20%. Details of these two datasets are summarized in TABLE I.

B. Experiment Setup

We design several experiments to evaluate the efficacy of our method for classifying anonymity network traffic. To answer RQ1, we evaluate the performance of the proposed raw traffic data processing method on different flow segmentation schemes and feature selection methods, and determine the optimal combination for subsequent experiments. To answer RQ2, we analyze the training process of RESGCN, and discuss the confusion matrix of the classification results on the test dataset. To answer RQ3, we compare the classification performance of RESGCN and the state of the art of traffic classification methods on the test dataset. It should be noted that both AnonymityNet and ISCXVPN2016 datasets are used to conduct the above evaluation experiments. To make the experimental evaluation clearer, we introduce the experimental environment and evaluation metrics as follows.

1) *Experimental Environment:* All the evaluations are conducted in Python 3.7 with the PyTorch framework of version 1.9.0 and running on the PC with Intel® Core™ i9-11900K@3.50 GHz, 64 GB RAM, and an NVIDIA GeForce RTX3090 GPU.

2) *Evaluation Metrics:* To measure the classification performance of our method, we calculate the number of True Positive (T_p), True Negative (T_n), False Positive (F_p), and False Negative (F_n). Based on the above definition, Recall, Precision and F_1 can be obtained:

$$Recall = \frac{T_p}{T_p + F_n}, \quad (11)$$

$$Precision = \frac{T_p}{T_p + F_p}, \quad (12)$$

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}. \quad (13)$$

Beside, Floating-Point-Operations (FLOPs) is used to evaluate the model complexity.

³SJTU-AN21 dataset: <https://github.com/iZRI/The-SJTU-AN21-Dataset>

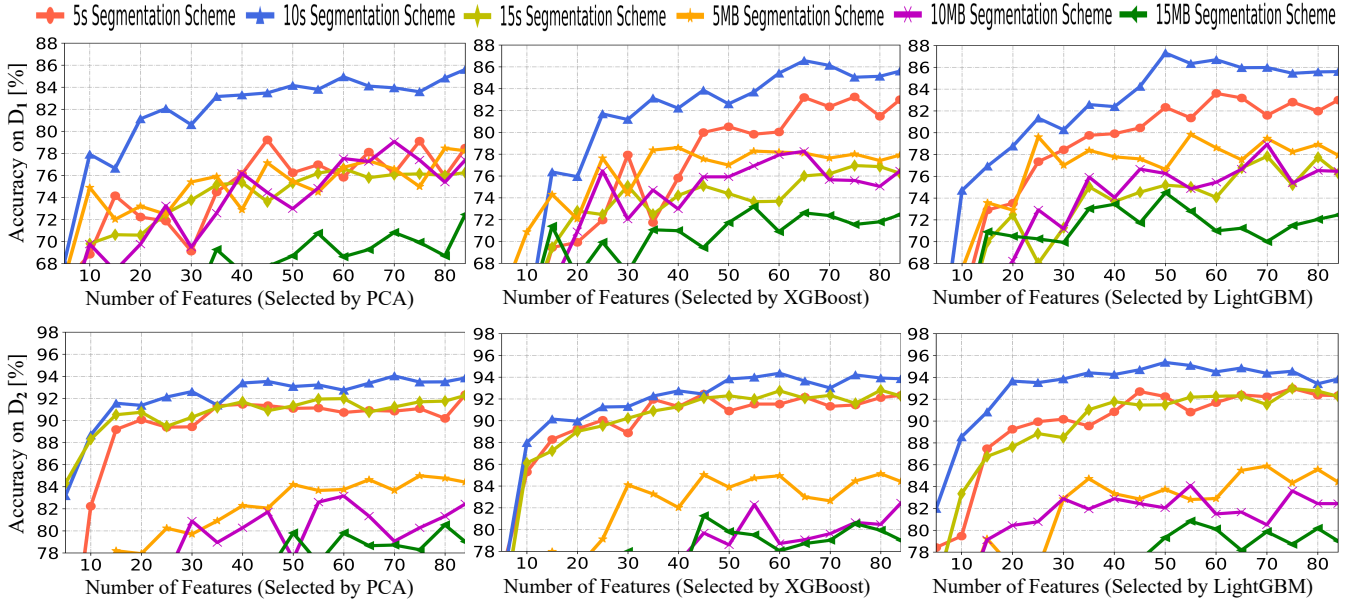


Fig. 5. Classification accuracy with different flow segmentation schemes and feature selection methods.

3) *Hyper-parameter Configurations*: For all the experiments in this section, we optimize the cross-entropy loss using stochastic gradient descent (SGD) optimizer over 100 epochs. The initial learning rate is set to 0.01 with the batch size 80 and a momentum of 0.9.

C. Efficacy of Raw Traffic Data Processing Scheme (to RQ1)

Our aim is to classify real-world anonymity network traffic effectively, so it is not enough to design the classifier. After capturing the raw traffic, different traffic data processing schemes can significantly affect the classification performance. In this stage, we conduct research on the following three main influencing factors: flow segmentation, feature combination, and flow sequence length.

We evaluate the performance of 6 segmentation schemes (i.e., time-based 5s, 10s, 15s, and size-based 5MB, 10MB, and 15MB) and 3 feature selection methods (i.e., PCA-based, XGBoost-based, and LightGBM-based). XGBoost and LightGBM are both GBDT-based ML algorithms, which calculate the gain of the j -th feature in the tree to reflect the feature importance. The PCA algorithm reduces the data dimension by maximizing the variance of the target dimension. We use the training dataset for feature selection and evaluate the accuracy of our method on the test dataset.

It can be seen from Fig. 5 that the 10s feature segmentation scheme achieves the best classification performance on both datasets, and the 15MB segmentation scheme has the worst performance. Furthermore, through data analysis of the 15MB flow segmentation scheme, it is found that the duration of many flows using this segmentation scheme is very long, and it is evident that the spatio-temporal correlation between flows is weakened. Besides, in the comparison of feature selection algorithms, the LightGBM-based method and the XGBoost-based method achieve higher maximum accuracy (MaxAcc) than the PCA-based method. However, the performance of

TABLE II
PERFORMANCE OF THREE FEATURE SELECTION METHODS.

Performance	PCA	XGBoost	LightGBM
MaxAcc ₁ [FN]	85.60% [84]	86.56% [65]	87.31% [50]
MaxAcc ₂ [FN]	94.03% [70]	94.36% [60]	95.37% [50]
Time (s)	4.17	62.9	11.11

¹ MaxAcc₁ and MaxAcc₂ represent the maximum accuracy on AnonymityNe dataset and ISCXVPN2016 dataset, respectively.

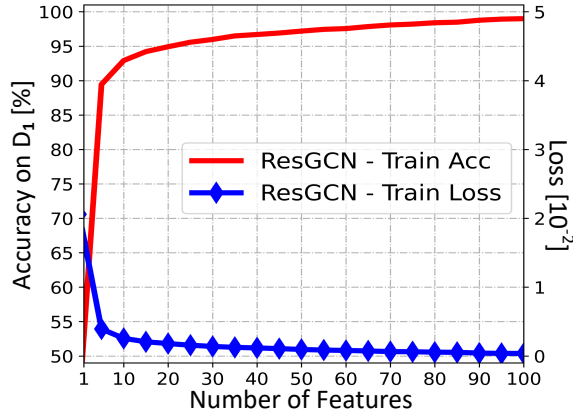
² FN represents the number of features with the maximum accuracy.

the PCA-based method is better when the number of features is small, which is because the PCA method maps the original high-dimensional features to low-dimensional, so very low-dimensionality can retain more information. TABLE II comprehensively shows the performance of the three feature selection methods. LightGBM-based feature selection method achieves the highest MaxAcc on two datasets, and the number of features is 50. PCA method has achieved apparent advantages in speed, but it is not satisfactory in terms of accuracy. It can be concluded that the LightGBM-based feature selection method has the best overall performance because of its high accuracy and computational efficiency. Thus, we use the LightGBM-based method to set the feature number to 50 with the LightGBM for subsequent experiments.

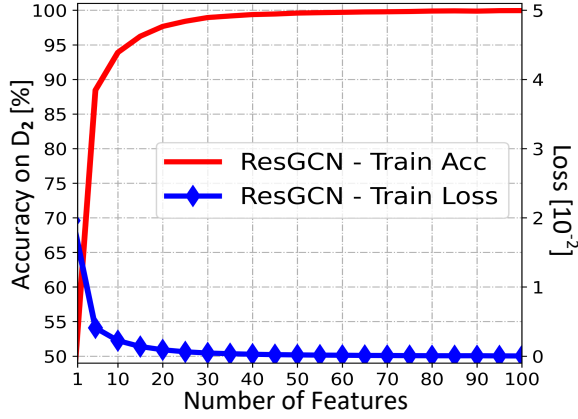
D. Performance of RESGCN Classifier (to RQ2)

The train accuracy and loss using the RESGCN model for the two datasets are shown in Fig. 6. It can be seen that the RESGCN model learns the feature information of the training dataset very well during the training process, which has satisfactory performance in terms of accuracy and loss.

To comprehensively evaluate the classification performance of RESGCN, we analyze the confusion matrices of the classification results on the test datasets. As shown in Fig. 7,



(a) Training process on D_1

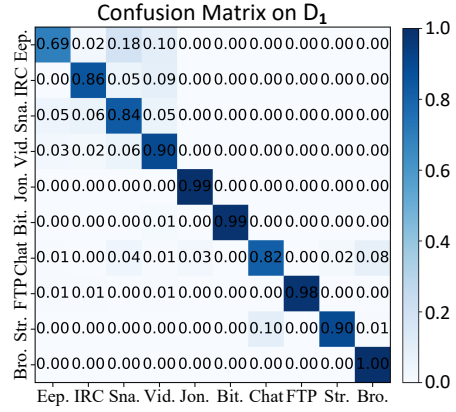


(b) Training process on D_2

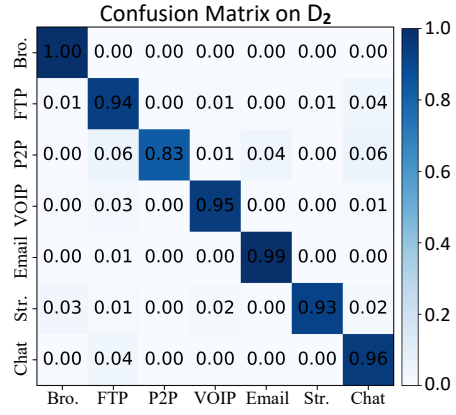
Fig. 6. Train accuracy and loss of RESGCN model on two datasets.

our classifier can effectively classify the traffic generated by different services. On the AnonymityNet dataset, the traffic of the three types of anonymity networks (i.e., I2P, Tor, JonDoNym) is correctly distinguished. In terms of identifying anonymity network services, services on I2P networks are the most difficult to identify, and there have been some misclassifications. On the ISCXPVN2016 dataset, we can see that the classifier has some classification errors between the *FTP* and *Chat* service traffic. By referring to the official description of the ISCXPVN2016 dataset, we found that both services contain the traffic generated by the Skype application, which is the main cause of misclassification.

To further examine the level of benefit that each component of RESGCN brings to the performance, an ablation study is performed on RESGCN. The evaluation results are reported in TABLE III. We also present the results of RESCNN as a baseline, which uses residual CNN blocks with similar structures to residual GCN blocks. First, the removal of GFI leads to significant performance drops, which illustrates the necessity of GFI for information interaction between generated features. Compared with RESGCN, RESGCN w/o RFI is much less powerful on two datasets. Thus it could be concluded that RFI contributes to RESGCN to a considerable extent since



(a) Confusion matrix of D_1



(b) Confusion matrix of D_2

Fig. 7. Performance confusion matrices of classifier on two datasets.

RFI leverages the relationship graph of the flow sequence for more reasonable and effective feature extraction. Moreover, the performance of RESGCN w/o RES (i.e., preserving GCN units, but without residual structure) shows that the residual structure can effectively avoid network degradation through direct signal transmission. The experimental results of RESGCN w/o GCN and the RESCNN both prove that the GCN unit plays a vital role in feature extraction of the flow sequence.

E. Comparison with Other Methods (to RQ3)

To comprehensively evaluate our model, we compare RESGCN with a range of baselines and state-of-the-art models, as listed below:

- **NB**, **SVM**, and **C4.5** are implemented based on the popular data mining tool *Weka* [30].
- **CNN** and **LSTM** models use the structure similar to RESGCN (i.e., the same number of layers and input/output size), and use flow sequences as input for classification.
- **2D-CNN** proposed by Wang *et al.* [9] directly reads traffic pcap files and converts them into grey images (size of each image is 28×28) for classification.
- **3D-CNN** is an improved model of the 2D-CNN proposed by Zhang *et al.* [8], which uses multiple channels to

TABLE III
ABLATION STUDY.

Model	AnonymityNet		ISCXVPN2016	
	Acc.	F_1	Acc.	F_1
RESGCN	87.3%	87.4%	95.4%	95.4%
ResCNN	82.1%	81.6%	91.0%	91.0%
RESGCN w/o GFI	79.4%	78.7%	91.8%	91.9%
RESGCN w/o RFI	81.9%	81.6%	88.9%	89.1%
RESGCN w/o GCN	78.1%	77.6%	87.1%	87.3%
RESGCN w/o RES	81.0%	80.6%	92.4%	92.5%

TABLE IV
THE PERFORMANCE COMPARISON WITH OTHER METHODS.

Model	AnonymityNet		ISCXVPN2016		FLOPs (10^6)	Params (10^4)	Size (KB)	Speed (flows/ms)
	Acc.	F_1	Acc.	F_1				
NB	34.0%	27.0%	54.7%	57.2%	N/A	N/A	N/A	18.3
SVM	46.8%	42.8%	71.8%	72.0%	N/A	N/A	N/A	56.8
C4.5	48.4%	48.7%	78.5%	76.2%	N/A	N/A	N/A	113.6
CNN	78.3%	77.0%	90.2%	90.4%	46.5	16.2	642	84.1
LSTM	79.1%	77.9%	90.2%	90.3%	114.6	26.5	1,043	23.5
2D-CNN	65.0%	66.9%	86.2%	86.1%	1110.7	327.5	12,796	4.9
3D-CNN	71.9%	73.2%	87.6%	87.9%	113.4	44.7	1,749	8.8
LAttn	81.5%	81.1%	92.3%	92.3%	28.3	12.2	484	17.7
FS-Net	83.7%	86.3%	92.4%	92.6%	497.6	64.6	2,532	2.1
Ours	87.3%	87.4%	95.4%	95.4%	20.9	12.1	482	100.7

¹ We highlight the best in • and the worst in •.

² Speed is evaluated on CPU mode.

³ Since the three ML-based models are directly implemented and tested on the data mining tool *Weka*, the results of FLOPs, Params, and model size cannot be obtained (marked N/A).

enrich feature information during image conversion. The model input is the 3D tensor with the size of $22 \times 22 \times 3$, which can be visualized as 24-bit RGB images.

- **LSTM+Attn** proposed by Yao *et al.* [17] uses LSTM combined with a self-attention mechanism to pay attention to important flows in the flow sequence.
- **FS-Net** is a LSTM-based network proposed by Liu *et al.* [13] for encrypted traffic classification. This method leveraged reconstruction loss to ensure that the extracted features by LSTM contain more information of the raw flow sequences.

TABLE IV shows the results of traffic classification performance. It can be seen that RESGCN achieves the best performance in both two datasets in terms of almost all evaluation metrics. We can observe that the classification results of traditional ML-based methods are usually not ideal, demonstrating those methods' limited abilities to classify complex network traffic. The classification performance of the 2D-CNN and 3D-CNN models (i.e., directly reading pcap files without calculating the statistical features; see Section III-A) is very limited. CNN, LSTM and LDAE all use statistical features and the flow sequence to achieve significant performance improvements. However, due to the lack of mining the inherent relationship of the flow sequence, they still cannot achieve high accuracy. The LAttn model learns the inherent relationship of the flow sequence through the attention mechanism,

and its performance is further improved. FS-Net achieves a better feature representation of encrypted traffic by introducing reconstruction loss, effectively improving classification performance. However, this model structure also brings a larger amount of parameters. Our RESGCN designs the model structure from a brand-new perspective. It leverages generated relationship graphs for feature extraction of flow sequence, which significantly improves the classification performance on anonymity network traffic.

Further, we analyze the model complexity, model parameter size, model size, and speed. These evaluations are very important for some deployments, where a much smaller model (or faster model that can analyze traffic in real-time) is more important than the performance classification of a model that, in the end, cannot be run on some devices. Since the Memory and CPU usage are easily interfered with by other programs, we use FLOPs to reflect the model complexity for evaluating the hardware consumption of the running model. The model parameter size also affects memory usage during model inference. Besides, the model size indicates the disk space occupied by this model, and the speed indicates the number of flows that the model can process per millisecond. We observe that, due to the use of 2D convolution for feature extraction, both the 2D-CNN and 3D-CNN models are highly complex. The complicated gate structure of the LSTM-based model leads to very low computational efficiency. Thus, the

speed of the three LSTM-based models (i.e., LSTM, LAttn, and FS-Net) is very slow. Benefiting from the effective feature extraction of flow sequences by GCN, RESGCN can achieve accurate classification without large parameters. We also noticed that the C4.5 has a fast speed, but the subpar classification performance limits its deployment. It can be concluded that the RESGCN classifier achieves the highest classification accuracy as well as has low complexity and high speed due to excellent structural design.

VI. CONCLUSION

In this paper, we propose a novel flow sequence-based network traffic identification framework, which leverages RESGCN to exploit attribute relationships and time relationships between flows and successfully identifies different anonymity network services. Moreover, as an end-to-end and real-time traffic identification method, our framework can effectively process real-world traffic. It takes into account flow segmentation, which is used to generate and enrich the flow features from the raw traffic, and LightGBM-based feature combination, which avoids the insignificant features from reducing model performance and efficiency. The results of our experiments show that the RESGCN classifier achieves the highest classification accuracy as well as has low complexity and high speed due to excellent structural design.

ACKNOWLEDGMENT

This work was supported by the Foundation Item: Cyber Security from the National Key Research and Development Program of Shanghai Jiao Tong University under Grant 2019QY0703. Yijun Wang is the corresponding author.

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